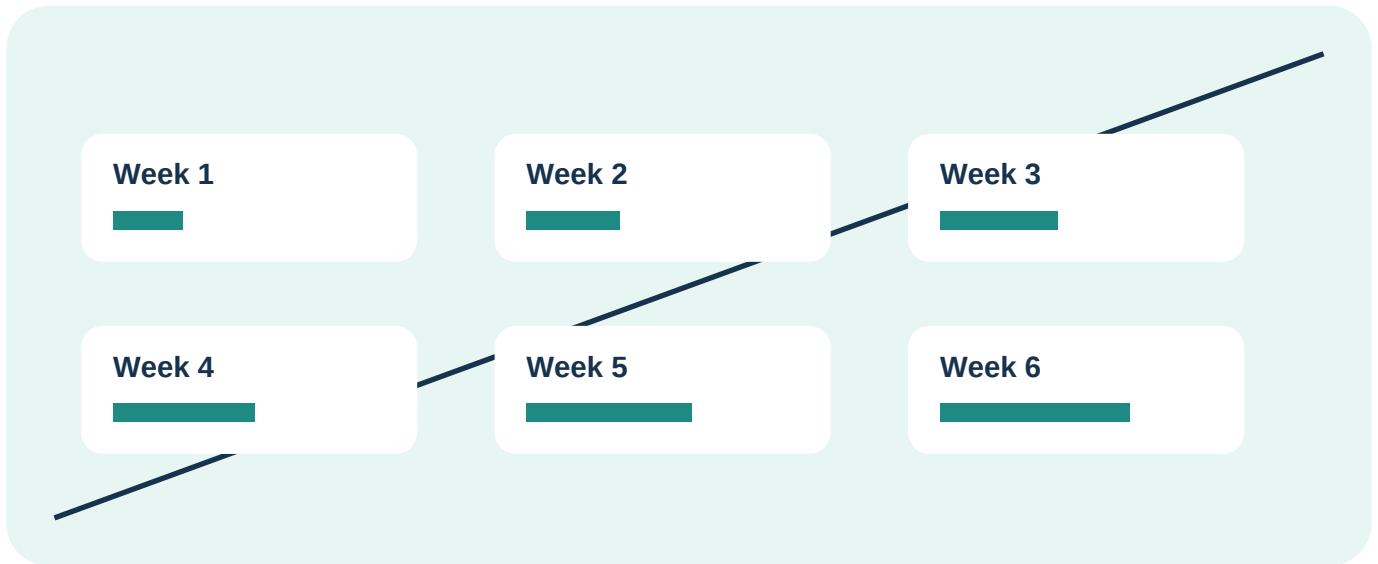


ANL559 Student Pre-Class Problem Briefs

Realistic weekly business scenarios for preparation before class.



Northstar Retail case sequence

Read each scenario, inspect the listed data files, and bring your own questions and assumptions to class.

Week 1 - Reporting Fire Drill

The Monday KPI pack is due, but the usual analyst is away.

?

Customers



\$?

Revenue

?

Tickets

KPI pack awaiting source checks

Stakeholder note

I need the weekly numbers by 10:00. I do not need perfect slides; I need numbers I can trust.

Problem Context

Northstar Retail sends a weekly KPI pack to the COO. The report is due every Monday morning, but the current process depends on manual copying from several source files.

The usual analyst is unavailable this week. The COO still needs a clear view of customer, revenue, and support performance.

People In The Room

- COO
- Analytics team
- Customer support lead

Data Available

File or material	What it contains
week1_customers.csv	Customer records, subscription status, plan type, login date, and paid status.
week1_sales.csv	June order records, gross amount, and refund amount.
week1_support.csv	Support ticket records, open/closed dates, and severity.
week1_metric_definitions.md	Business definitions for the weekly KPI pack.

Fields To Notice

- customer_id, subscription_status, plan, last_login_date, is_paid
- order_id, customer_id, order_date, gross_amount, refund_amount
- ticket_id, customer_id, opened_date, closed_date, severity

Bring To Class

- The Week 1 data files.
- One or two questions about field meanings.
- A short list of the KPI numbers you think the COO would expect.

Pre-Class Thinking

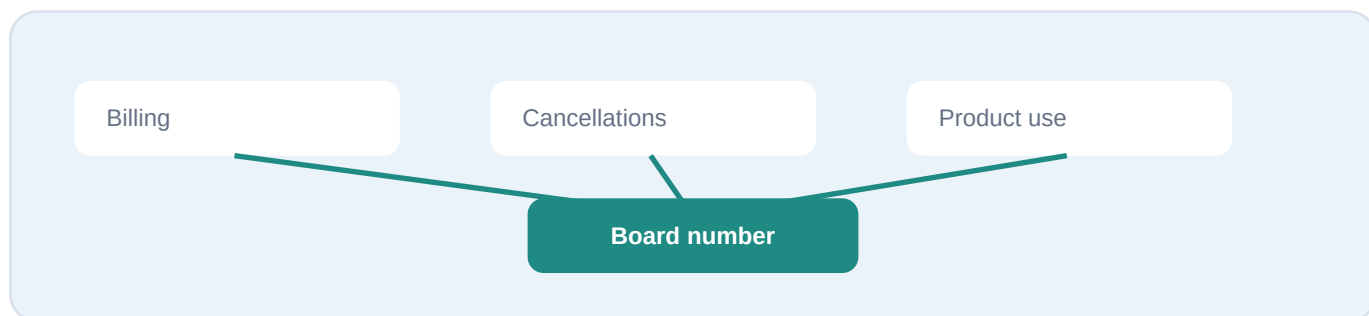
1. Which numbers would you expect in a weekly KPI pack for the COO?
2. Which data file would you use for each number?

3. Which words or business terms might need clarification before the report is sent?
4. What would make you confident that a number in the report is traceable?
5. What questions would you ask the COO before treating the report as final?

Pre-class notes

Week 2 - The Board Number Is Wrong

Finance and Product both have churn numbers in the board pack, and they do not match.



Stakeholder note

Before this goes to the board, tell me which number we should use and what it actually means.

Problem Context

Northstar Retail is preparing board materials. Finance and Product have reported different churn numbers, and both teams believe their number is reasonable.

The board paper needs one number that can be explained clearly. The team must understand what each source represents before choosing what to report.

People In The Room

- Finance lead
- Product lead
- Board secretary

Data Available

File or material	What it contains
week2_subscriptions.csv	Subscription plan, billing status, billing start date, and billing end date.
week2_cancellations.csv	Cancellation request date, billing end date, and cancellation reason.
week2_product_events.csv	Recent product activity by customer, including sessions and active feature count.

Fields To Notice

- customer_id, plan, billing_status, billing_start, billing_end
- customer_id, cancel_request_date, billing_end_date, cancel_reason
- customer_id, last_event_date, sessions_30d, active_feature_count

Bring To Class

- The three Week 2 data files.
- A short description of what each file appears to represent.
- Two questions you would ask Finance or Product before using the number.

Pre-Class Thinking

1. What might Finance mean by churn?
2. What might Product mean by churn or customer activity?
3. Which source would you expect each team to trust most?

4. What information would you want before choosing one number for the board paper?
5. What could make two reasonable teams report different numbers?

Pre-class notes

Week 3 - Limited Retention Budget

The business can contact only a limited number of customers.

All customers

Customers needing review

Limited contact capacity

Stakeholder note

If we can contact only a small group, who should we prioritise and why?

Problem Context

Northstar Retail wants to reduce customer churn, but the business cannot offer every customer a retention incentive.

Customer Success has limited time, and each offer has a cost. The business needs a practical way to decide which customers should be reviewed first.

People In The Room

- Customer Success manager
- Finance lead
- Retention programme owner

Data Available

File or material	What it contains
week3_retention_training.csv	Customer-level data about tenure, monthly fee, usage, support tickets, plan type, discounts, offer response, and next-month retention.

Fields To Notice

- customer_id, tenure_months, monthly_fee, usage_sessions_30d
- support_tickets_90d, plan_type, discount_last_90d
- post_offer_response, retained_next_month

Bring To Class

- The Week 3 data file.
- One simple customer-prioritisation idea.
- Two questions you would ask Customer Success before acting on the data.

Pre-Class Thinking

1. What does the business need to decide?
2. Which fields might indicate that a customer needs attention?
3. Which fields might indicate that a customer is valuable to retain?
4. What would make a simple prioritisation rule acceptable to a business stakeholder?
5. What could go wrong if the business contacts the wrong customers?

Pre-class notes

Week 4 - From Notebook To Tool

Operations wants something usable, not a notebook that only one analyst can run.

User input

Internal tool

Recommendation

Stakeholder note

Can someone on my team use this safely without asking the analyst to run it?

Problem Context

Northstar Retail has analysis that may help Customer Success prioritise customers.

Operations is interested, but they do not want to depend on an analyst manually running a notebook each time a decision is needed.

People In The Room

- Operations manager
- Customer Success team
- Analytics team

Data Available

File or material	What it contains
Week 3 decision context	Customer-prioritisation logic or notes from the previous week.
Suggested request format	Customer ID, tenure, fee, usage, support tickets, and plan type.

Fields To Notice

- customer_id, tenure_months, monthly_fee
- usage_sessions_30d, support_tickets_90d, plan_type

Bring To Class

- Your Week 3 prioritisation idea or notes.
- A sketch of what an operations user would enter.
- A sketch of what the user should receive back.
- Two questions you would ask Operations before releasing an internal tool.

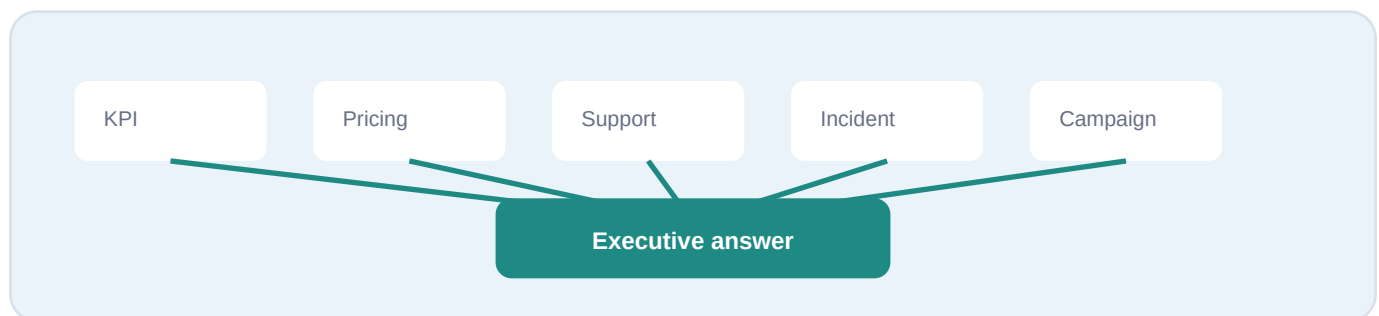
Pre-Class Thinking

1. Who would use this internal tool or process?
2. What information would the user need to provide?
3. What output would be useful to the user?
4. What could make a tool confusing or risky for an operations user?
5. What instructions would someone need if the original analyst is unavailable?

Pre-class notes

Week 5 - The Executive Question

The CEO wants to know why churn changed in June.



Stakeholder note

Give me the short version, but make sure we are not overstating the evidence.

Problem Context

The CEO asks why churn changed in June. A useful answer must be clear, evidence-based, and careful about uncertainty. Northstar has several internal notes and summaries that may help answer the question.

People In The Room

- CEO
- Analytics lead
- Customer Success lead
- Product lead

Data Available

File or material	What it contains
week5_knowledge_base.md	Short internal source notes about churn, pricing, support, product incidents, and marketing activity.

Fields To Notice

- KPI Pack
- Pricing Note
- Support Summary
- Product Incident Note
- Marketing Campaign Note

Bring To Class

- The Week 5 source note.
- A short draft answer to: Why did churn rise in June?
- Two places where you think the available evidence is strong or weak.

Pre-Class Thinking

1. What does the CEO likely want to know?
2. Which source notes seem most relevant to the question?
3. What would make an executive answer convincing?

4. What would make an executive answer risky?
5. What follow-up analysis might be needed before making a stronger claim?

Pre-class notes

Week 6 - Launch Review

The launch committee wants a recommendation, not a feature tour.



Evidence

Scope

Owner

Launch recommendation

Stakeholder note

What can go live, what should wait, and who owns it if something goes wrong?

Problem Context

Northstar Retail is considering whether the analytics work from the previous weeks is ready for use beyond the classroom setting.

The launch committee wants to know what is ready, what is not ready, what evidence exists, and who would own follow-up actions.

People In The Room

- Launch committee
- Analytics lead
- Operations manager
- Customer Success manager
- Executive sponsor

Data Available

File or material	What it contains
Week 1 reporting work	KPI reporting artefact, definitions, and checks.
Week 2 reconciliation work	Notes on conflicting numbers and source comparison.
Week 3 retention work	Customer-prioritisation or decision notes.
Week 4 internal tool work	Notes on user inputs, outputs, and operating needs.
Week 5 executive answer work	Evidence-based answer and source notes.

Fields To Notice

- Weekly artefacts and notes
- Evidence gathered across the course
- Questions, risks, and unresolved decisions

Bring To Class

- Your weekly artefacts or notes from Weeks 1-5.
- One claim you feel confident defending.
- One claim you are not yet confident defending.
- Two questions you expect the launch committee to ask.

Pre-Class Thinking

1. Which part of your work is most ready to use?
2. Which part still needs more evidence?
3. Who would own the work if it were used after class?
4. What should stop the team from releasing something too broadly?
5. What would you say if a stakeholder asks for a stronger claim than the evidence supports?

Pre-class notes
